
Aplicacions Distribuïdes

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(*) Some slides come from other sources

Web applications

- Applications that will be accessed through the Internet by a browser
- The ones developed in class
 - Based on the Servlet and Java Server Pages specification originally defined by Sun, now owned by Oracle
 - Other: asp Microsoft, Python, php, scripts cgi, javascript, etc.

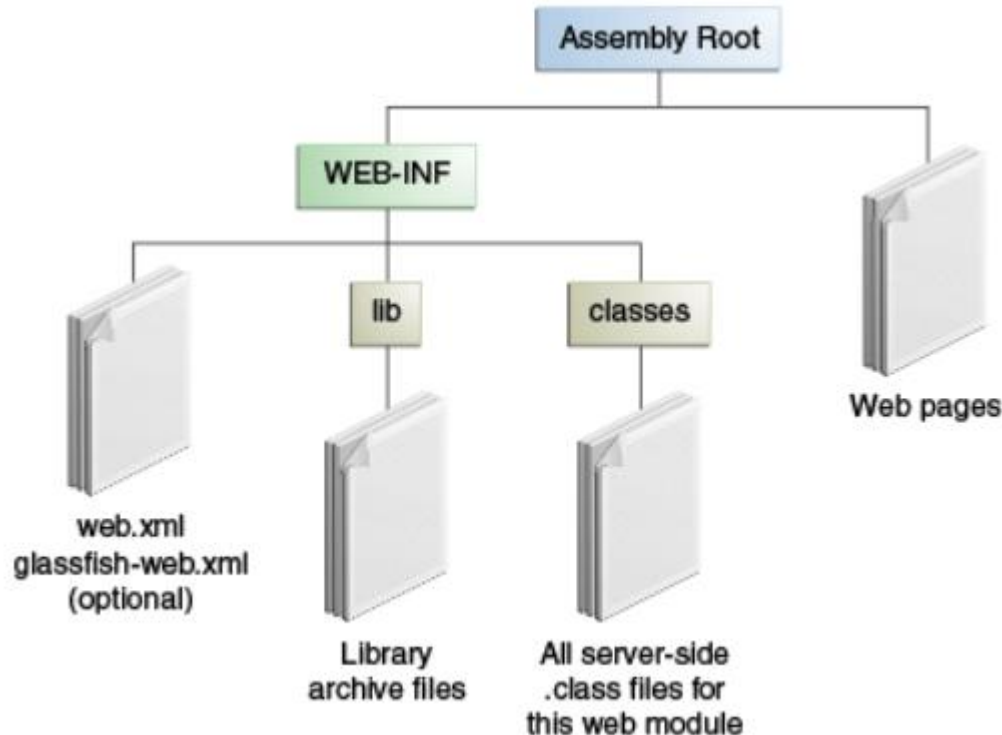
Static Web Applications

- HTML pages
- Images
- Static resources
 - Text files
 - Pdf files
 - Other static files ...
- Elements do not change over time
 - Good to cache them (proxy, browser, etc.)

Dynamic Web Applications

- Generate HTML code using some programming language
 - Oracle: Java Server Pages, Servlets
 - Microsoft: ASP, C#, .net
 - Javascript
 - PHP
 - Executables (.exe, scripts, etc.)
 - Python
 - Any software that generates HTML code
- Elements can be dynamically generated
 - Database access, files and external services, etc.
 - Also static elements (images, documents, etc.)

Deployed J2EE web applications structure



Web pages folder

- Static HTML pages
- Java Server Pages
- Images or other public files (pdf, doc's, etc.)
- It may have subfolders

Web applications structure

Source: <https://javaee.github.io/tutorial/packaging003.html>

Structure (continued)

- **WEB-INF folder**
 - Special directory of the web application, which contains configuration information
 - classes
 - Contains java classes (.class) that uses the application
 - lib
 - Contains java libraries (.jar) that uses the application
 - web.xml
 - Web application configuration file
 - Now you can also use Java Annotations (@)

Reference: <https://docs.oracle.com/javase/tutorial/java/annotations/basics.html>

web.xml example file (J2EE 9)

```
<?xml version="1.0" encoding="UTF-8"?>
```

```
<web-app xmlns="https://jakarta.ee/xml/ns/jakartaee"  
  xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"  
  xsi:schemaLocation="https://jakarta.ee/xml/ns/jakartaee  
    https://jakarta.ee/xml/ns/jakartaee/web-app 5 0.xsd"  
  version="5.0">
```

XML Schema

```
  <session-config>  
    <session-timeout>  
      30  
    </session-timeout>  
  </session-config>
```

```
</web-app>
```

Reference: <https://jakarta.ee/specifications/servlet/5.0/jakarta-servlet-spec-5.0.pdf>

Servlets

- Servlet
 - Class dynamically loaded in the server to add functionality
 - Execution in a Java Virtual Machine (JVM)
 - Platform independent (!!)
 - How it responds to the client?
 - Generates HTML code
 - Redirects to another page
 - jsp, static html, another servlet, etc.

Servlet lifecycle

- 1) Create and initialize servlet
- 2) Process web clients requests
- 3) Destroy the servlet, freeing memory and resources (*garbage collection*)

API Servlets

- 2 java packages
 - jakarta.servlet, javax.servlet until J2EE 8
 - jakarta.servlet.http, javax.servlet.http until J2EE 8
- Generic servlets (ja*.servlet)
 - Do not have **main()**
 - Invoke **service()** method when receiving a request

HTTP Servlets

- Specific for HTTP
- **service()** method already implemented
 - Invokes appropriate HTTP method
 - doXxx (where Xxx can be Get, Post, other HTTP methods)

Java Server Pages (JSP)

- Mix HTML and Java code
- They are especially used to build interfaces (not command processing)
- Special tags to integrate HTML code with Java code (`<% %>`)

Servlets vs Java Server Pages

- Similar functionality
- Jsp, easier installation and development
 - Only the user interface, more complex to process
- Server internally converts JSP into a servlet
 - Generates Java code to build the response
 - Everything that is HTML code is generated and sent to the client using Java code
 - Compiles the code
 - Invokes this code when the jsp is called

Servlets vs Java Server Pages

- jsp compiled class is stored in a server internal folder
 - IDE's have improved and now servlets generated from jsp are easier to reach when debugging

Application Servers

- The ones in the subject are based on the J2EE model
 - Originally defined by Sun, then Oracle, now supported by Apache and Eclipse
- They work both with web applications and web services

Application Servers

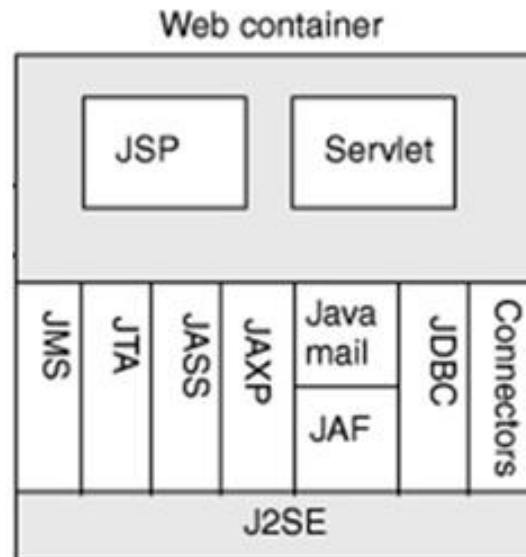
- They provide functionality to support the implementation of complex web applications and services (also used by companies)
- J2EE applications are portable between different application servers
 - Developed in Java language
 - Predefined folder structure
 - .war files, Web Application Archive

Reference: Compatible implementations with J2EE

<https://www.oracle.com/technetwork/java/javaee/overview/compatibility-jsp-136984.html>

<https://jakarta.ee/compatibility/>

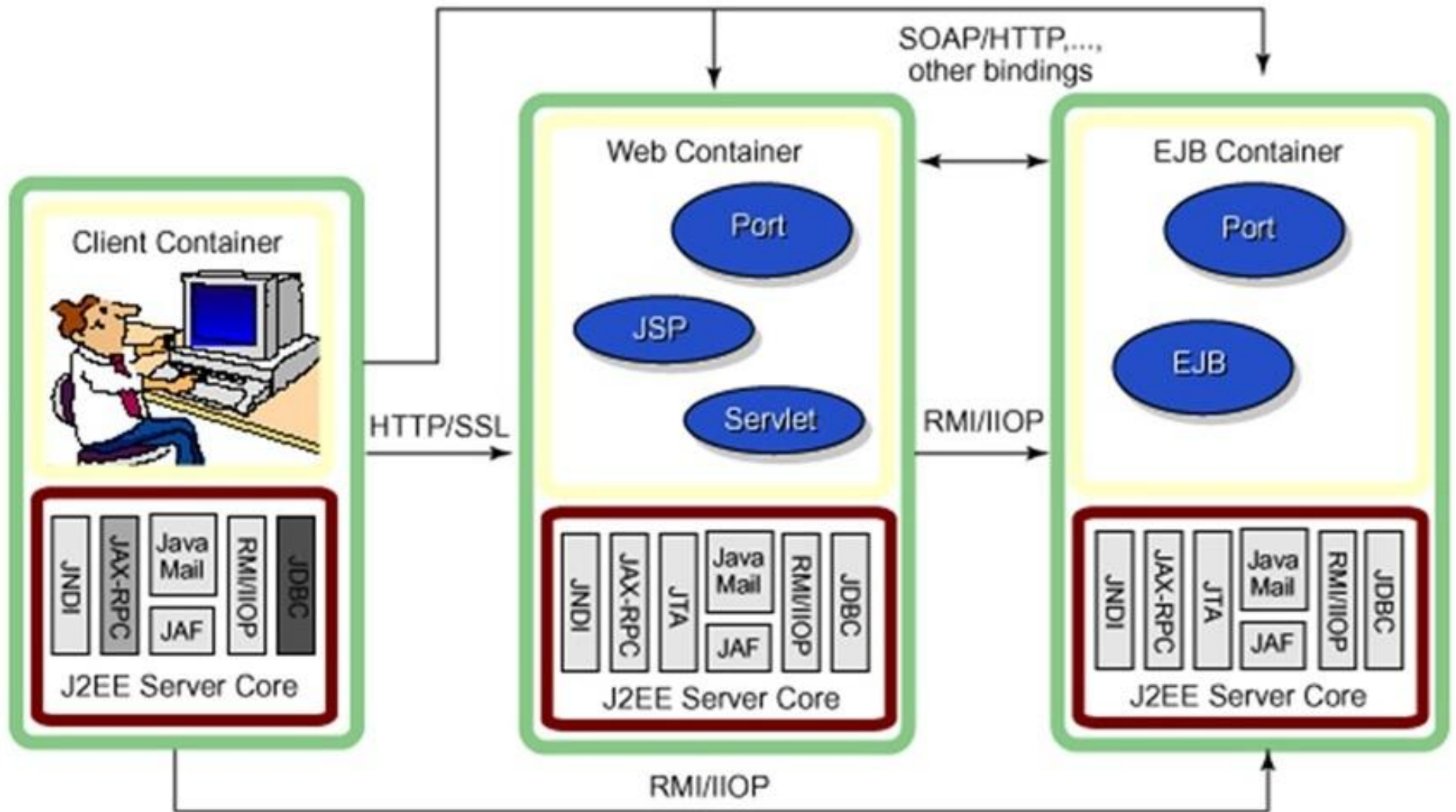
J2EE Application Server Architecture



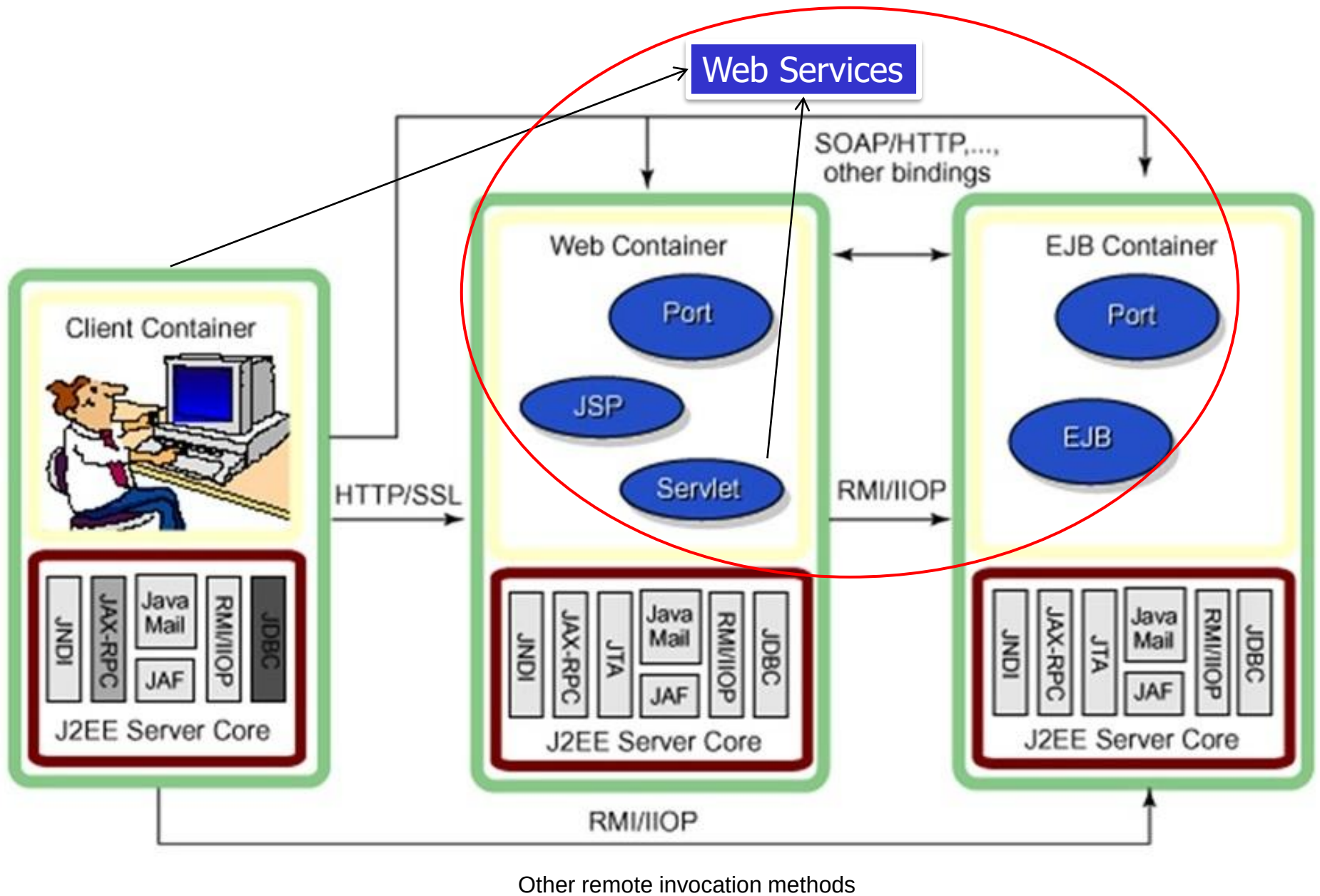
J2SE = Java 2 Platform, Standard Edition
JAAS = Java Authentication and Authorization Service
JAF = JavaBeans Activation Framework
JAXP = Java API for XML Parsing

JDBC = Java Database Connectivity
JMS = Java Message Service
JSP = Java Server Page
JTA = Java Transaction API

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Reference: <https://www.semanticscholar.org/paper/J2EE-Architecture-and-Patterns-in-Enterprise-Guo/89e570fa9317c99e950219eeff1cb9091063af45>



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